

Customer Churn Prediction and Retention Intelligence System

End-to-End Machine Learning Project
Internship Evaluation Submission

Dataset: Telco Customer Churn (Kaggle)

Source: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

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1. Abstract

This project presents a complete end-to-end machine learning system for predicting customer churn in a telecommunications company. Using the publicly available Telco Customer Churn dataset from Kaggle (7,043 customers, 21 features), the system implements data cleaning, exploratory data analysis, business KPI calculation, feature engineering, multiple classification models, hyperparameter tuning, SHAP explainability, customer risk segmentation, and a rule-based business recommendation engine. The final XGBoost (Tuned) model achieves 84.71% ROC-AUC and 81.02% Recall on the test set. All results are presented through a professional interactive Streamlit dashboard with 8 pages covering executive overview, churn analysis, model performance, explainability, risk prediction, high-risk customers, business insights, and dataset exploration.

2. Introduction

Customer churn — the phenomenon of customers discontinuing their relationship with a business — is a critical challenge for telecommunications companies. Acquiring new customers costs significantly more than retaining existing ones, making churn prediction and proactive retention essential for sustainable business growth.

This project builds a comprehensive Customer Churn Prediction and Retention Intelligence System that goes beyond simple classification. It connects the entire analytics pipeline: DATA → CLEANING → EDA → KPIs → ML MODEL → MODEL EXPLANATION → CUSTOMER RISK → BUSINESS INSIGHTS → ACTIONABLE RECOMMENDATIONS.

The system is designed to help business stakeholders understand churn patterns, identify at-risk customers, and take data-driven retention actions.

3. Business Problem

The business wants to identify customers who are at risk of leaving and answer:

- How many customers are represented in the dataset?
- What is the overall churn rate?
- Which customer groups have higher churn?
- What factors are associated with churn?
- Which customers are predicted to be at high risk?
- What are the major churn drivers?
- Which high-risk customers have high business value?
- What actions could the business take to reduce churn?

The project focuses on turning machine-learning predictions into actionable business insights.

4. Objectives

- Clean and preprocess the Telco Customer Churn dataset without data leakage
- Perform comprehensive exploratory data analysis with data-driven insights
- Calculate meaningful business KPIs
- Build and compare multiple classification models (Logistic Regression, Random Forest, Gradient Boosting, XGBoost)
- Perform hyperparameter tuning for the strongest candidate model
- Apply SHAP explainability to the final model
- Generate customer-level risk scores and segmentation
- Create a rule-based business recommendation engine
- Build a professional interactive Streamlit dashboard
- Document all findings in a reproducible project format

5. Dataset Description

The dataset used is the Telco Customer Churn dataset from Kaggle, originally provided by IBM as a sample dataset.

Source URL: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

Property	Value
Rows	7,043
Columns	21
Duplicate Rows	0
Missing Values	11 blank strings in TotalCharges
Target Column	Churn (Yes/No)
Target Distribution	No: 5,174 (73.46%) Yes: 1,869 (26.54%)
Identifier	customerID (excluded from modelling)

The dataset contains 15 categorical features, 4 numerical features, 1 identifier column, and 1 target column.

6. Data Preprocessing

The following preprocessing steps were applied:

Step	Details
Duplicate Detection	0 duplicate rows found; no removal needed.
Whitespace Stripping	Applied str.strip() to all string columns.
TotalCharges Conversion	11 blank strings (customers with tenure=0) were replaced with 0.0. Column converted from string to

	float64.
Target Encoding	Churn: Yes → 1, No → 0, stored in 'Churn_Binary'. Original column preserved.
SeniorCitizen	Already binary (0/1); no conversion needed.
Missing Values	No remaining NaN values after TotalCharges treatment.
Categorical Encoding	OneHotEncoder applied within sklearn Pipeline (handle_unknown='ignore').
Numerical Scaling	StandardScaler applied within sklearn Pipeline.
Data Leakage Prevention	All transformations applied via ColumnTransformer fitted only on training data.

7. Exploratory Data Analysis

Comprehensive EDA was performed to understand churn patterns across multiple dimensions:

Analysis	Finding
Overall Churn	26.54% churn rate (1,869 out of 7,043)
Contract Type	Month-to-month: 42.7% churn One year: 11.3% Two year: 2.8%
Internet Service	Fiber optic: 41.9% DSL: 19.0% No internet: 7.4%
Payment Method	Electronic check: 45.3% (highest) Mailed check: 19.1% Bank transfer: 16.7% Credit card: 15.2%
Tenure	Churned customers avg ~18 months vs retained ~38 months
Monthly Charges	Churned customers pay ~\$74/mo vs retained ~\$61/mo
Senior Citizens	Higher churn rate than non-seniors
Tech Support/Security	Customers without these services churn at significantly higher rates
Paperless Billing	Paperless billing customers churn at higher rates

All EDA visualisations are interactive Plotly charts accessible through the Streamlit dashboard. Charts include churn distribution donut, horizontal bar charts for categorical variables, box plots for numerical variables, scatter plots for tenure vs charges, and correlation heatmaps.

[INSERT SCREENSHOT: Dashboard EDA Page showing churn analysis charts]

8. KPI Analysis

The following business KPIs were calculated from the dataset:

KPI	Value	Calculation
Total Customers	7,043	Count of all records in the dataset
Churned Customers	1,869	Customers with Churn = Yes
Retained Customers	5,174	Customers with Churn = No
Churn Rate	26.54%	Churned / Total × 100
Average Tenure	32.4 months	Mean of tenure column
Average Monthly Charges	\$64.76	Mean of MonthlyCharges
Month-to-Month Customers	~55%	Proportion on MTM contracts
Revenue at Risk (estimate)	Sum of MonthlyCharges for high-risk customers	Proxy metric

9. Feature Engineering

Six additional features were engineered from existing columns. All features use only pre-prediction information to avoid target leakage.

Feature	Description
TenureGroup	Binned tenure into 4 cohorts (0-12, 13-24, 25-48, 49+ months)
ServiceCount	Total number of active services subscribed
ChargePerService	MonthlyCharges / ServiceCount
IsMonthToMonth	Binary: 1 if contract is Month-to-month
HasProtection	Binary: 1 if TechSupport or OnlineSecurity = Yes
IsElectronicCheck	Binary: 1 if PaymentMethod = Electronic check

10. Machine Learning Methodology

A scikit-learn Pipeline architecture was used to ensure clean separation of preprocessing and modelling, and to prevent data leakage.

Pipeline Structure:

- ColumnTransformer: StandardScaler for 9 numerical features, OneHotEncoder for 15 categorical features
- Classifier: Logistic Regression, Random Forest, Gradient Boosting, XGBoost, XGBoost (Tuned)

Training Setup:

- Stratified train/test split: 80% training / 20% test
- 5-fold stratified cross-validation
- Class imbalance handling: class_weight='balanced' (LR, RF), scale_pos_weight (XGBoost)
- Random seed: 42 (reproducibility)
- No information from test set used during preprocessing or model selection

11. Model Comparison

All models were evaluated on the held-out 20% test set:

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV-AUC
Logistic Regression	74.24%	50.96%	78.34%	61.75%	84.20%	84.56%
Random Forest	77.71%	60.27%	47.06%	52.85%	82.12%	82.51%
Gradient Boosting	79.21%	63.11%	52.14%	57.10%	83.52%	83.81%
XGBoost	75.51%	52.64%	77.27%	62.62%	83.72%	83.87%
XGBoost (Tuned)	75.66%	52.70%	81.02%	63.86%	84.71%	84.83%

[INSERT SCREENSHOT: Dashboard Model Performance page showing comparison table and ROC curves]

12. Hyperparameter Tuning

Hyperparameter tuning was performed on XGBoost using RandomizedSearchCV with 30 iterations, 5-fold stratified cross-validation, and ROC-AUC as the scoring metric.

Search Space: n_estimators [100, 200, 300], max_depth [3, 4, 5, 6], learning_rate [0.05, 0.1, 0.15], subsample [0.8, 0.9, 1.0], colsample_bytree [0.8, 0.9, 1.0], min_child_weight [1, 3, 5]

Parameter	Best Value
subsample	0.9
n_estimators	100
min_child_weight	1
max_depth	4
learning_rate	0.05
colsample_bytree	0.9

Result: Tuned XGBoost achieved 84.83% CV-AUC and 84.71% test ROC-AUC, improving over the base XGBoost (83.72% test AUC). The tuned model was included in the final comparison and selected as the final model.

13. Final Model

XGBoost (Tuned) was selected as the final model based on:

- Highest ROC-AUC (84.71%) — Best threshold-independent discrimination
- Highest Recall (81.02%) — Catches the most churners (minimises false negatives)
- Best F1 (63.86%) — Best balance of precision and recall

For churn prediction, Recall is prioritised over Accuracy because missing a churner (false negative) has greater business cost than a false alarm (false positive). A customer predicted as at-risk who was actually going to stay incurs only the cost of a retention offer, whereas missing a churner results in permanent revenue loss.

14. Model Evaluation

Final model (XGBoost Tuned) performance on the test set:

Metric	Value
Accuracy	75.66%
Precision	52.70%
Recall	81.02%
F1-Score	63.86%
ROC-AUC	84.71%
CV-AUC (5-fold)	84.83%

The confusion matrix and ROC curve are available in the Streamlit dashboard on the Model Performance page.

These metrics were generated from the actual model execution on the actual dataset. No metrics were fabricated or manually entered.

[INSERT SCREENSHOT: Confusion matrix and ROC curve from dashboard]

15. Explainability (SHAP)

SHAP (SHapley Additive exPlanations) was applied to the final XGBoost model to provide both global and individual-level explanations.

Global Feature Importance: A SHAP summary plot shows which features contribute most to churn predictions across all customers. Top features include contract type, tenure, internet service, monthly charges, and payment method.

Individual Explanation: For any selected customer, SHAP values explain which features pushed the prediction toward (positive SHAP) or away from (negative SHAP) churn.

Important: SHAP values describe feature contributions to model predictions. They do not imply causation. Language used: 'contributes to prediction' / 'associated with prediction'.

[INSERT SCREENSHOT: SHAP summary plot and individual explanation from dashboard]

16. Customer Risk Segmentation

For each customer, the final model generates:

- Churn probability (0.0 to 1.0)
- Predicted churn class (Yes/No)
- Risk category (Low / Medium / High)

Risk Category Thresholds:

- Low Risk: Churn probability < 0.35
- Medium Risk: $0.35 \leq \text{Churn probability} < 0.65$
- High Risk: Churn probability ≥ 0.65

These thresholds are configurable in the application configuration section. The customer risk table includes customerID, tenure, contract, monthly charges, churn probability, predicted churn, risk category, and recommended action.

17. Dashboard

The Streamlit dashboard consists of 8 interactive pages:

1. Executive Overview — KPI cards, churn distribution, top drivers, key insights
2. Customer & Churn Analysis — Interactive filters, segmented charts, scatter plots
3. ML Model Performance — Model comparison, ROC curves, confusion matrices, hyperparameter tuning
4. SHAP Explainability — Global importance, individual customer explanations
5. Customer Risk Prediction — Interactive form for real-time predictions
6. High-Risk Customers — Filterable risk table, CSV downloads
7. Business Insights — FACT → INSIGHT → RISK → OPPORTUNITY → ACTION framework
8. Dataset Explorer — Data overview, sample viewer, cleaning log

The dashboard uses a dark theme with professional styling, KPI cards with gradient effects, Inter font family, and responsive layout.

[INSERT SCREENSHOT: Executive Overview page of the dashboard]

[INSERT SCREENSHOT: Customer Risk Prediction form]

18. Business Insights

Insight Area	Finding
Contract Type	Month-to-month customers churn at 42.7% vs 11.3% (one-year) and 2.8% (two-year). Contract type is the strongest churn signal.
Fiber Optic	Fiber optic customers churn at 41.9% vs 19.0% for DSL, despite higher investment.
Electronic Check	Electronic check users churn at 45.3% — highest across all payment methods.
New Customers	Churned customers average ~18 months tenure vs ~38 months for retained. Early engagement is critical.
Monthly Charges	Churned customers pay higher monthly charges (~\$74 vs ~\$61).
Protective Services	Customers without TechSupport or OnlineSecurity have substantially higher churn.
High-Value at Risk	Customers with high charges AND high churn probability represent concentrated revenue exposure.
ML Early Warning	The model identifies 81% of churners before they leave.

19. Risks

- Large month-to-month cohort represents significant churn exposure
- Fiber optic customer churn creates disproportionate revenue impact
- Customer acquisition costs are wasted when new customers churn quickly
- High-churn payment segment reduces efficiency of blanket retention campaigns
- High-value customer loss creates outsized revenue impact

- Over-reliance on model without human review may miss contextual factors

20. Opportunities

- Contract conversions from month-to-month to longer-term can structurally reduce churn
- Resolving fibre service quality issues can improve satisfaction in high-value segments
- Encouraging auto-pay may correlate with improved retention
- Structured onboarding improves satisfaction and reduces early-stage churn
- Value perception improvements can reduce price-driven churn without price reductions
- Free trials of protective services can improve retention
- Personalised retention outreach is most cost-effective for high-value customers

21. Recommended Actions

Condition	Action
High risk + MTM contract	Offer incentivised annual or 2-year contract upgrade
High risk + Short tenure	Proactive onboarding check-in, welcome discount, dedicated support
High risk + High charges	Personalised bundle offers or loyalty discounts
High risk + Fiber optic	Investigate service quality; offer proactive technical support
High risk + Electronic check	Encourage auto-pay with a small incentive
High risk + High value	Escalate to retention team immediately
All customers	Score monthly; route high-risk to retention within 48 hours

Disclaimer: These are business suggestions based on observed data patterns, not guaranteed outcomes. Correlation does not imply causation. Recommendations should be validated against current organisational data before deployment.

22. Limitations

- The dataset is publicly available and results are specific to this Telco dataset
- The model is trained on a static snapshot; real-world deployment requires periodic retraining
- Correlation does not imply causation — feature importance describes statistical associations
- Predicted churn is not guaranteed churn — predictions are probabilistic estimates
- Revenue-at-risk calculations are estimates based on MonthlyCharges, not actual future losses
- The model uses only features in the dataset; enrichment with additional data could improve performance
- The model should be validated on current business data before real-world deployment

23. Future Scope

- Deep learning models for larger, more complex datasets
- Time-series churn prediction incorporating temporal behaviour patterns
- Customer Lifetime Value (CLV) integration for prioritisation
- A/B testing framework to measure retention intervention effectiveness
- Real-time scoring API for CRM integration
- NPS and support ticket data enrichment
- Automated MLOps retraining pipeline
- Fairness analysis to audit predictions for demographic bias

24. Conclusion

This project successfully demonstrates a complete end-to-end machine learning pipeline for customer churn prediction. The final XGBoost (Tuned) model achieves 84.71% ROC-AUC and identifies 81.02% of churners, providing a strong foundation for proactive retention strategies.

Key findings show that contract type, internet service, payment method, and tenure are the strongest factors associated with churn predictions. Month-to-month contract customers churn at 42.7%, fibre optic customers at 41.9%, and electronic check payers at 45.3%.

The interactive Streamlit dashboard makes these insights accessible to business stakeholders, enabling data-driven decision-making for customer retention. The recommendation engine provides specific, actionable suggestions based on each customer's risk profile.

The system is designed as a reproducible, professional project suitable for internship evaluation and demonstrates competency in data science, machine learning, business intelligence, and application development.

25. References

26. Telco Customer Churn Dataset — <https://www.kaggle.com/datasets/blatchar/telco-customer-churn>
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